Web Scraping: Data Gathering from Web

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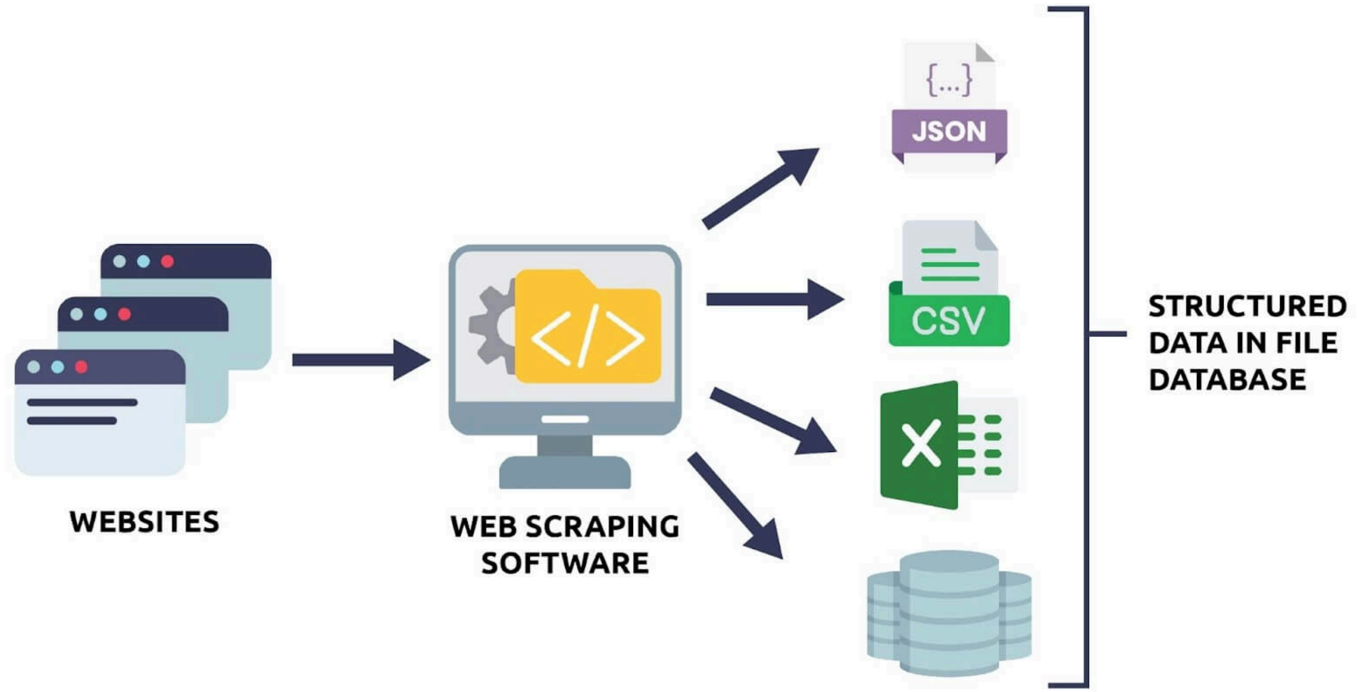
Outline

- What is Web Scraping?
- What is HTML?
- Preparation for Web Scrapping: Selenium and Chrome Driver

What is Web Scraping?

Web Scraping

- Process of *automatically* extracting specific data from a target website
- Usually done with a specific purpose of data analysis



Example: Develop a Price Tracking System

Data scraping is necessary to track the prices of the target products:

- Target website: Coupang
- Target data: Daily min/max prices

Applications of Web Scrapping

Businesses leverage web scraping tools to extract data from websites.

1. **E-commerce price tracking**: Scrape product names, prices, and ratings from sites like Amazon or eBay to monitor price changes over time.
2. **News monitoring**: Scrape headlines, article text, and publication dates from news websites to perform sentiment analysis or track trending topics.
3. **Social media analysis**: Extract posts, hashtags, likes, and comments from platforms like Twitter or Instagram for research or marketing insights.
4. **Market analysis**: Collect product prices, customer reviews, and ratings from online stores to analyze pricing trends, customer preferences, and competitor positioning.
5. **Stock price prediction**: Extract prices of target stocks from websites to develop and validate a stock price prediction model

Key steps typically include:

1. **Identify the target website** – Determine which website and which pages contain the data you need.
2. **Inspect the web page structure** – Use browser developer tools to understand the HTML elements where the data resides.
3. **Send requests to the website** – Programmatically fetch the web page content (often using HTTP requests).
4. **Parse the data** – Extract the relevant information from the HTML, JSON, or other formats (using tools like BeautifulSoup, lxml, or Selenium).
5. **Store the data** – Save the extracted data in a structured format like CSV, JSON, or a database.
6. **Respect legal and ethical considerations** – Check the website's terms of service, robots.txt, and avoid overloading the server.

What is HTML?

HTML (HyperText Markup Language)

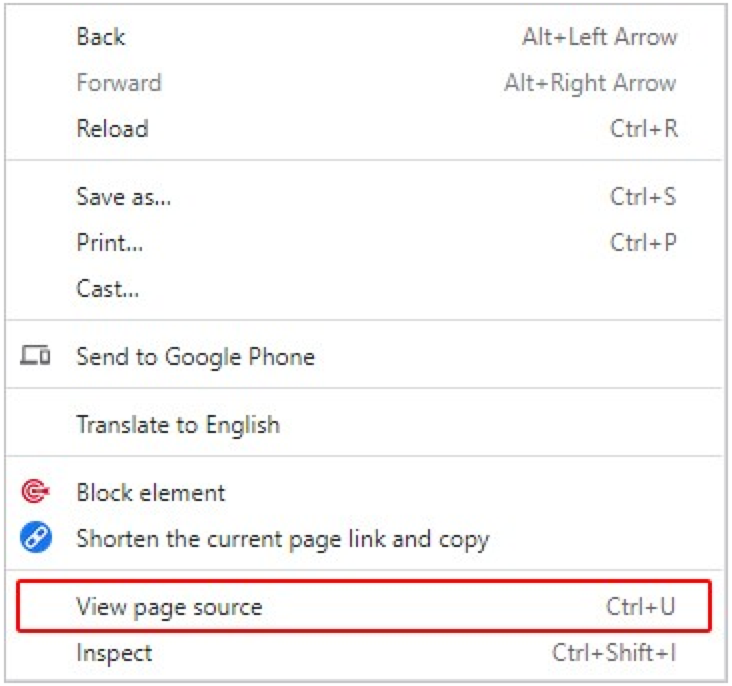
- Web pages have many visual and textual elements.
- HTML codes serve to position all of this content on web pages.

```
<!DOCTYPE html>
    <html>

        <head>
            <title> Title here </title>
        </head>

        <body>
            Web page content goes here.
        </body>

    </html>
```

How to check HTML source code

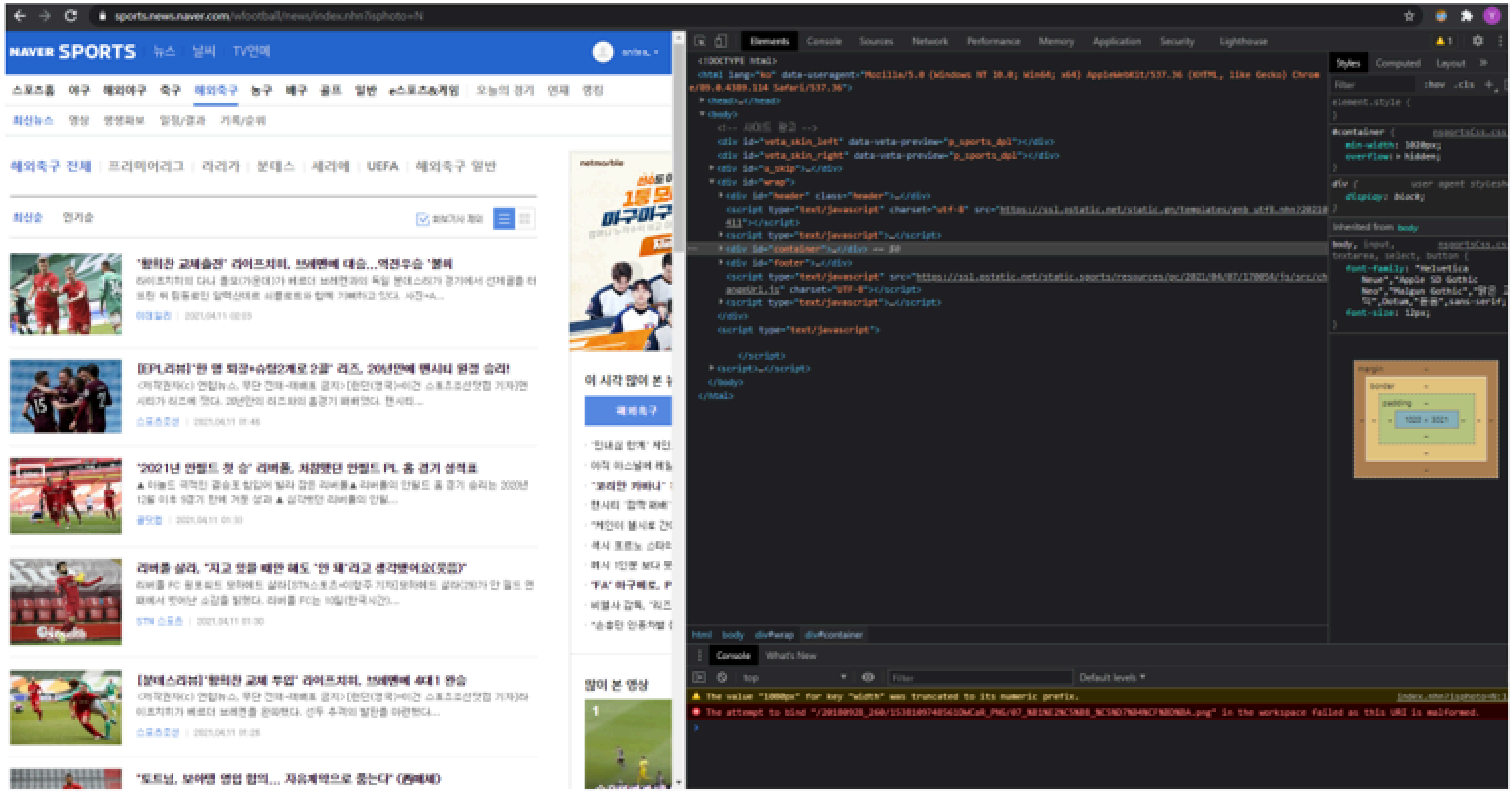
To see the HTML code of the webpage you are currently viewing,

- Place mouse cursor on an empty space on the page and right-click
- Select "View page source (페이지 소스 보기)" (Shortcut: `Ctrl` + `U`)



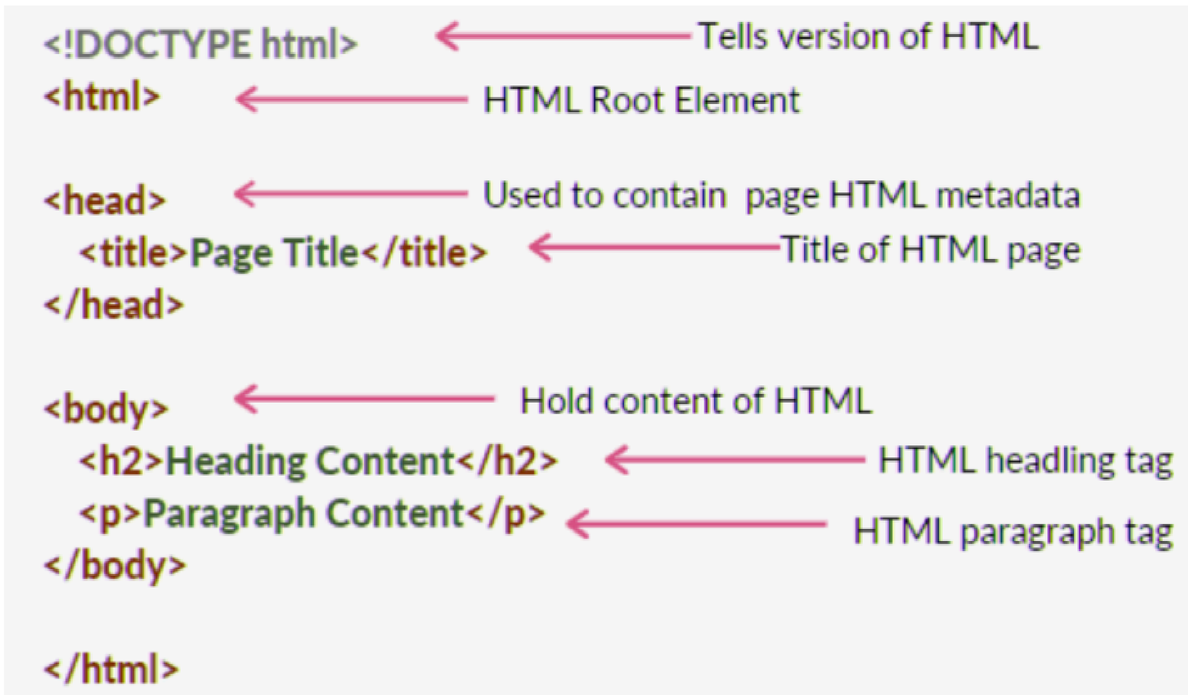
Or, you can use the **Inspect (검사)** menu for more functions:

- Place mouse cursor on an empty space on the page and right-click
- Select "Inspect (검사)" (Shortcut: `Ctrl` + `Shift` + `I`)



HTML Page Structure

It contains the essential two main building-blocks, `head` and `body`, upon which all web pages are created.



HTML Tag

- The actual keyword or name enclosed in angle brackets (`< >`)
- Tells the browser what kind of content to expect
- Example: <a> tag for links

> 📝 An element (e.g., `a href` attribute) is the complete structure, including the opening tag, content (if any), and the closing tag.

Table: HTML Tags

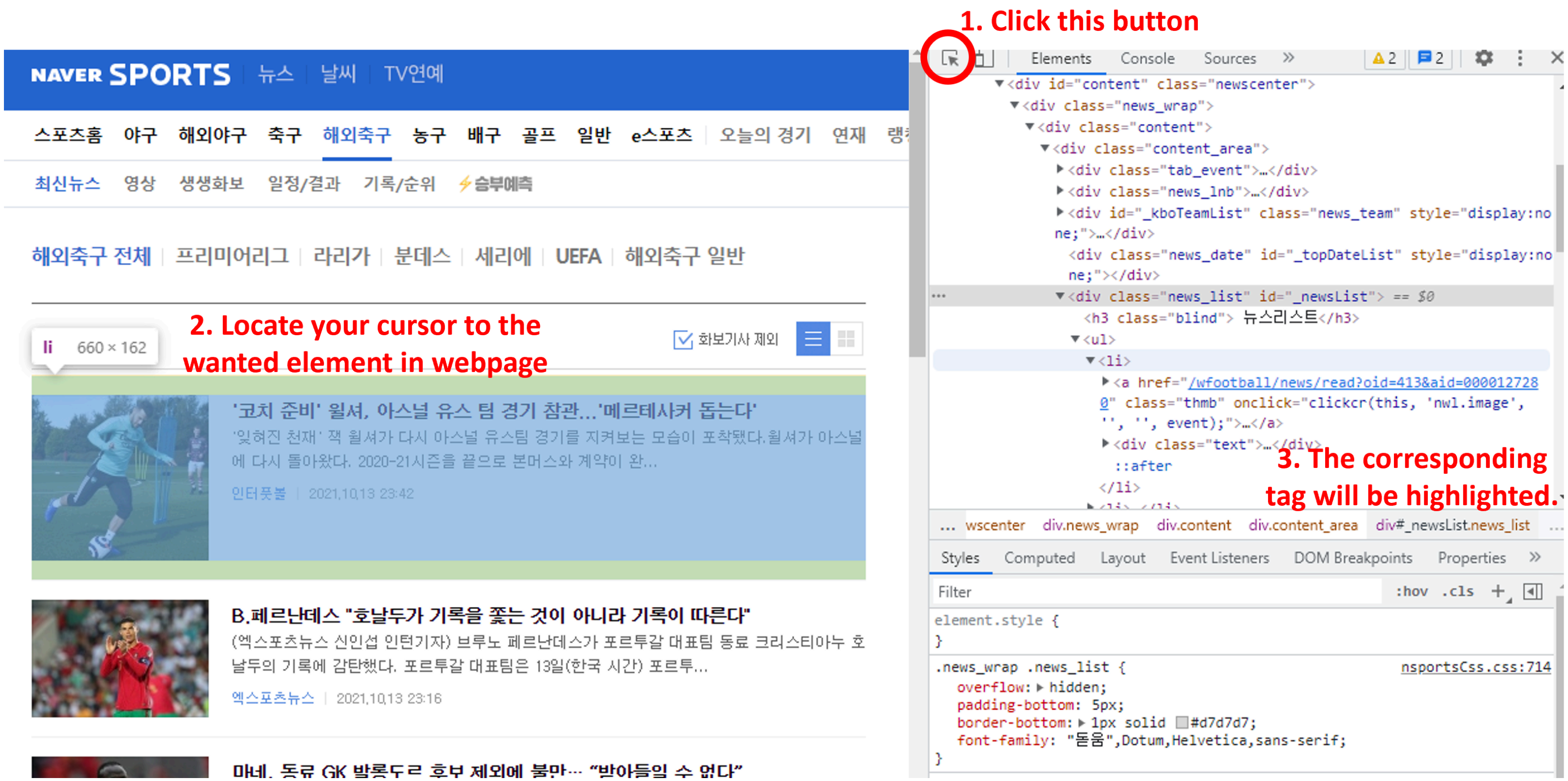
Tag	Description
`<html>`	The root element of an HTML document
`<head>`	Contains meta-information about the webpage
`<body>`	Contains the visible content of the webpage
`<h1>` to `<h6>`	Headings of various levels (`h1` being the largest)
`<p>`	Defines a paragraph
`<a>`	Defines a hyperlink
`<img>`	Embed an image
`<ul>`	Defines an unordered list
`<ol>`	Defines an ordered list
`<li>`	Defines a list item
`<table>`	Defines a table

Today's Menu: Daum Sports News

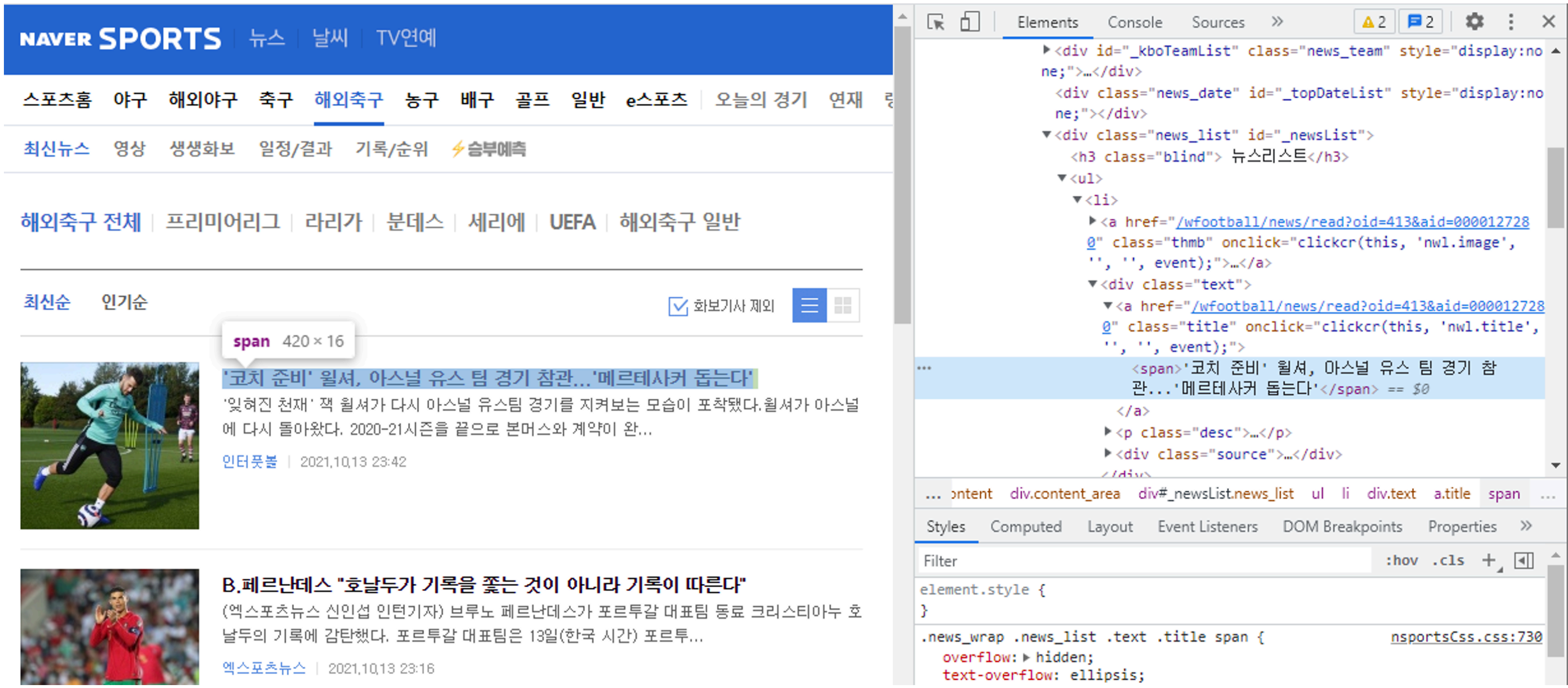
- Daum Sports News (https://sports.daum.net/worldsoccer/news/breaking)
- Scrapping attributes of all news in the first page of the **'global football news'** > **'news'** menu.
    - Title
    - News text
    - Publisher
    - Datetime

To scrap the data, you need to analyze the structure of the HTML code.

Finding the Wanted Tag



Moving cursor will change the location of the highlight!



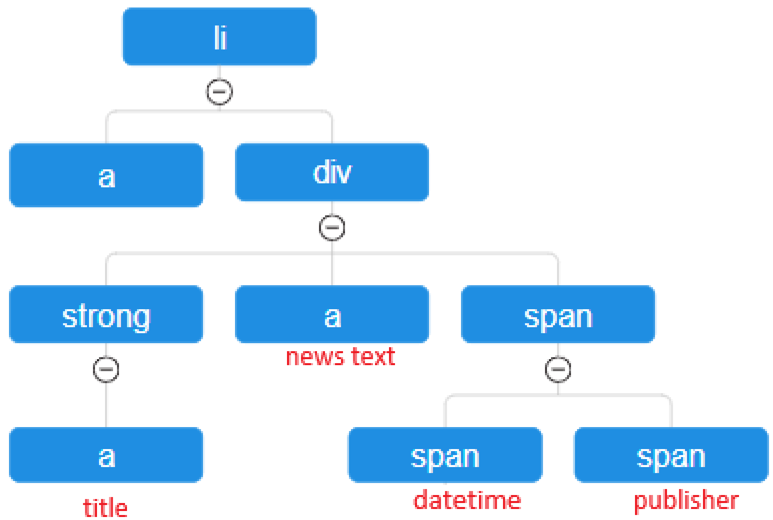
Findings

1. All news items in the page is contained in the `<ul class="list_news">` element. Each news item corresponds to a `<li>` element.

```
...
<ul class="list_news">
    <li data-searchid="313XXX"> ... </li>
    <li> ... </li>
    <li> ... </li>
    ...
    <li> ... </li>
</ul>
...
```

2. By analyze the tree structure of a `<li>` element, we can find the target data:

```
<li data-searchid="313XXX">
    <a href="https://v.daum.net/v/BYnlkg5rhX"><img src="//img1....."></a>
    <div class="wrap_cont">
        <strong class="tit_news">
            <a href="..." class="link_txt" ...> 맨유 드디어...</a>
        </strong>
        <a href="..." class="link_desc"> [스타뉴스 | 박건도 기자] 유럽 소식에...</a>
        <span class="info_news">
            <span class="txt_info txt_num">
                <span class="screen_out">작성날짜</span>2분 전
            </span>
            <span class="txt_info">스타뉴스</span>
        </span>
    </div>
</li>
```

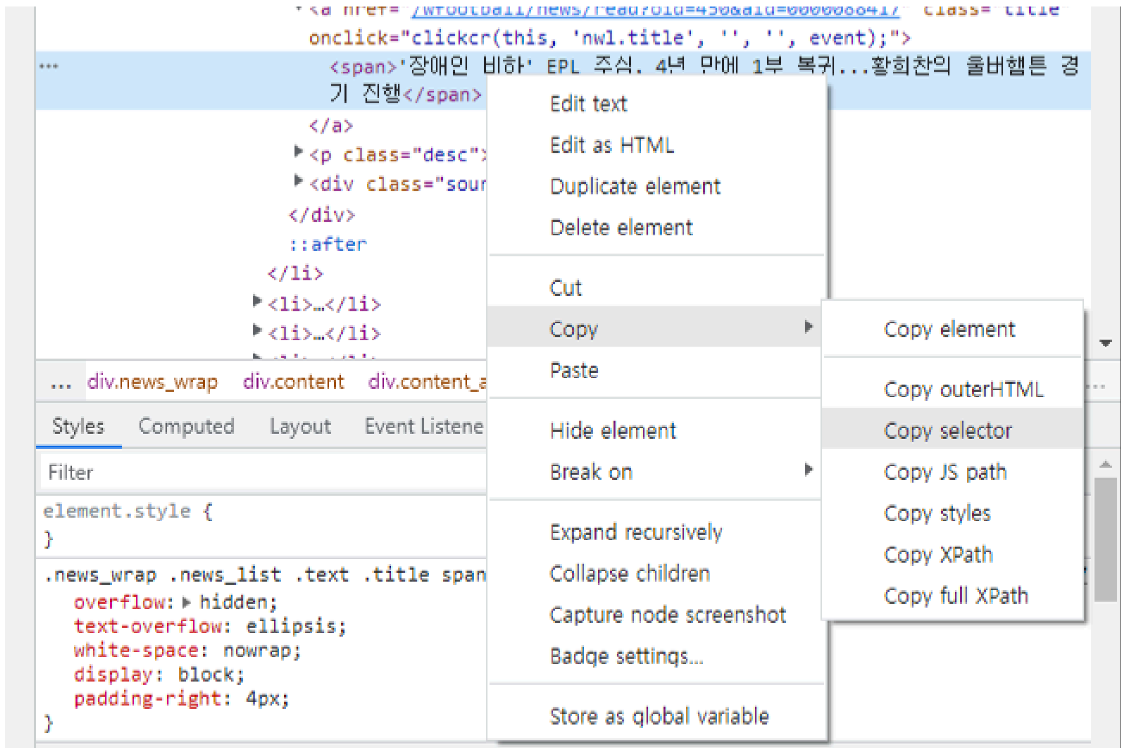


- Title: `li > div > strong > a`
- News text: `li > div > a`
- Publisher: `li > div > span > span`
- Datetime: `li > div > span > span:nth-child(2)`

> 📝 These patterns are called **CSS Selector**!
> A CSS selector is a pattern used to "find" or select HTML elements on a webpage

CSS Selector in Developer Tool in Chrome

You can copy the CSS selector pattern of the target element:



Preparation for Web Scraping

Preparation

- Chrome browser
  - We assume you already have the Chrome browser installed on your machine!
- **Selenium**
- **Chrome driver**

Selenium

An open-source library that is designed for **automating web browsers**.

- Can browse web like a human
- Can be used to scrapping information on web

How to install:

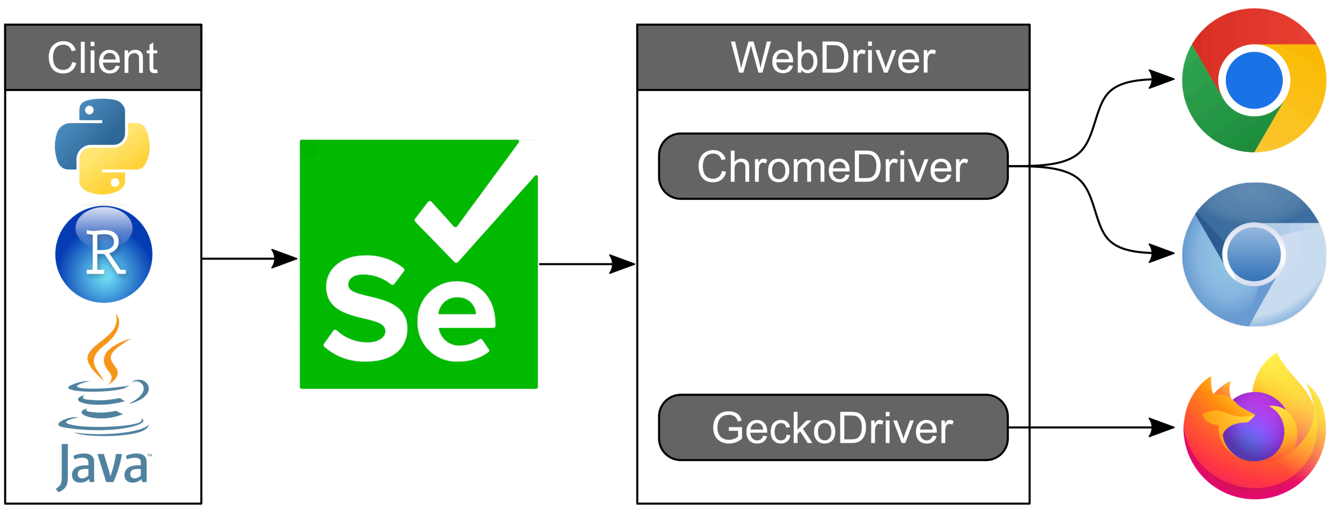
1. Open the Anaconda prompt
2. Enter the following:

```
pip install selenium
```

> 📝 If required, press `y` (yes).

Chrome Driver

A separate executable file that enables Selenium WebDriver to automate Google Chrome browsers.

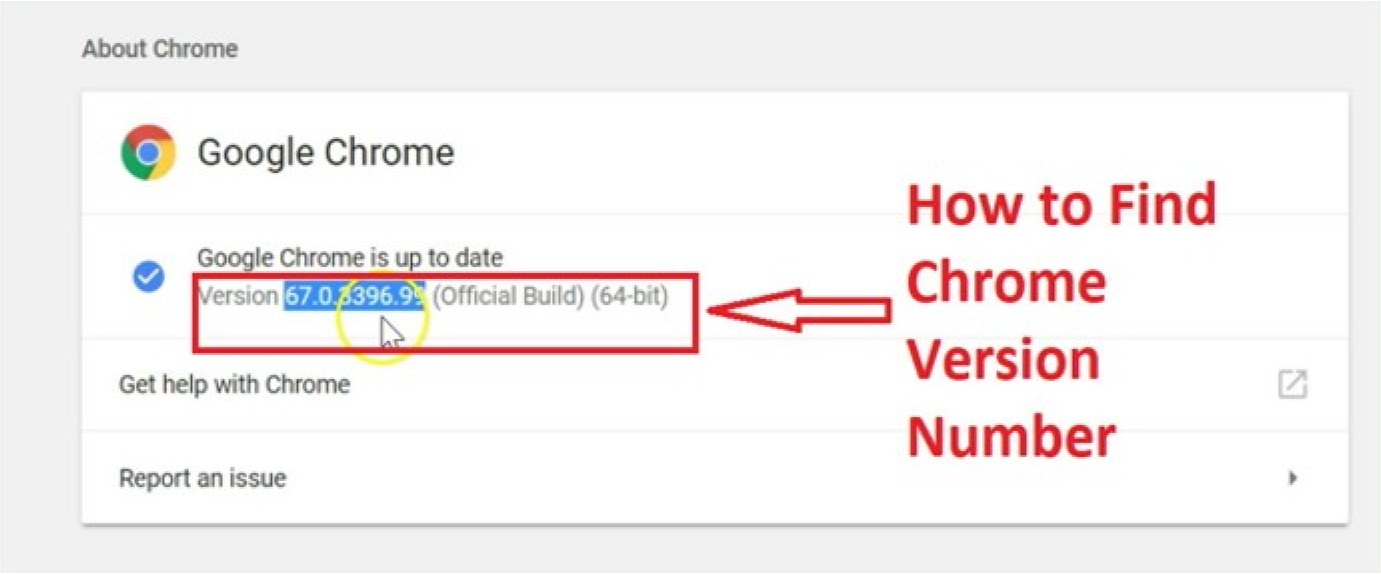


To use it, you need to **download a ChromeDriver version** that is **compatible with your Chrome browser version**.

How to Download Chrome Driver?

1. Update your Chrome browser to the latest version and check its version

- You can check the version by entering the following url in the address bar: chrome://settings/help



2. Download the Correct Chrome Driver

- Go to the official Chrome for Testing availability dashboard: https://googlechromelabs.github.io/chrome-for-testing/
- Find the version that matches the one you just checked.
- Download the chromedriver zip file for your operating system (e.g., `win64` for Windows, `mac-x64` for Mac).

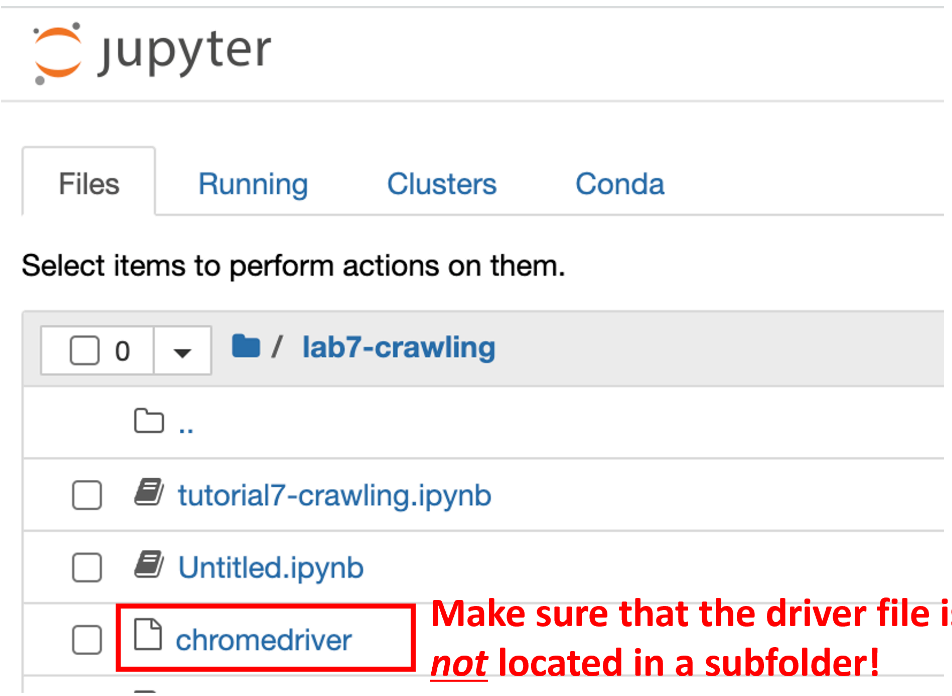
> 📝 If your Chrome browser version is lower than those in the dashboard, go to https://chromedriver.chromium.org/downloads

3. Unzip the zip file and you will get the driver file

- Windows: `chromedriver.exe`
- Mac: `chromedriver`

4. **Place the driver file in the same folder as the code file**

- e.g., the current Jupyter notebook file you are working on



Make sure that the driver file is *not* located in a subfolder!

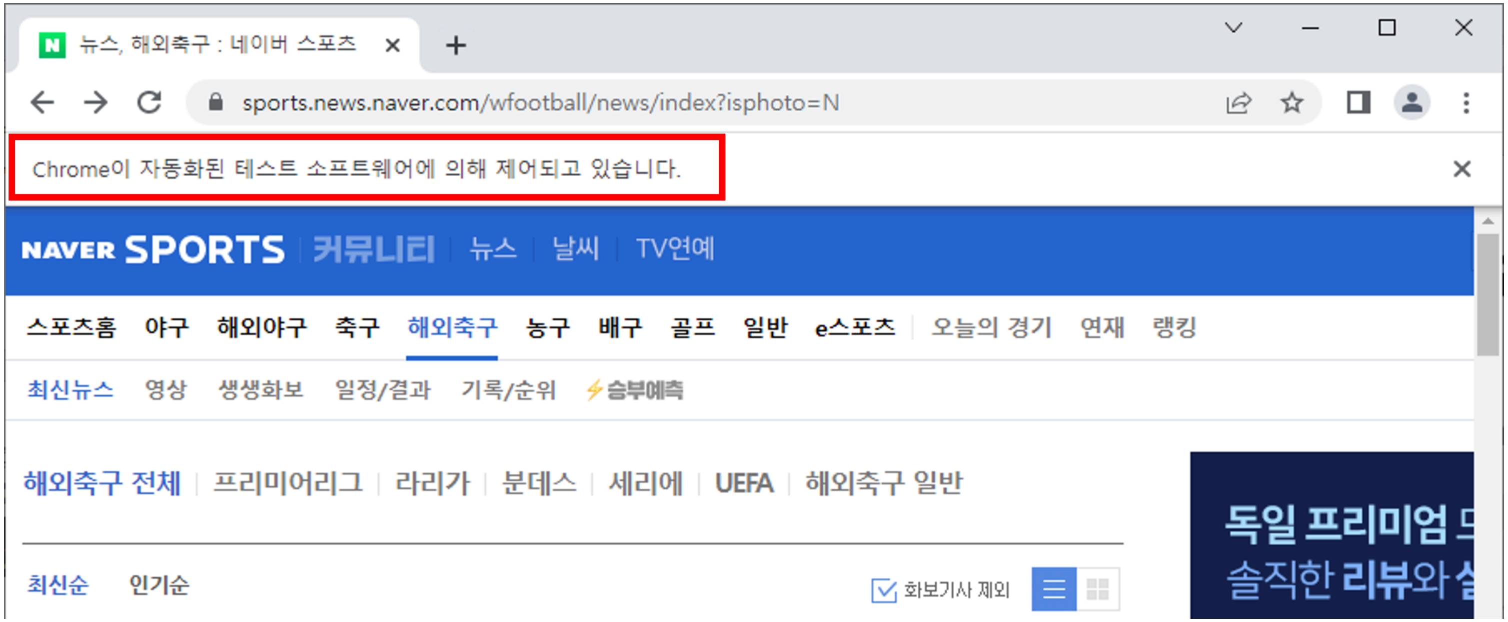
Using Chrome Driver on Python

The code below will open a new selenium browser window:

```
from selenium import webdriver

# set driver path depending on your OS
driver_path = './chromedriver.exe' # windows
driver_path = './chromedriver'     # mac

cService = webdriver.ChromeService(executable_path=driver_path)
driver = webdriver.Chrome(service=cService)
```

Selenium browser

Below the address bar, you will see the following text: `Chrome is being controlled by automated test software.`



> ⚠️ Don't close the opened browser until your web scraping process is completed!

[MacOS only] If you encounter the following error:

```
In [11]: from selenium import webdriver

         driver = webdriver.Chrome('/Users/ejk/Downloads/chromedriver')

/var/folders/78/m1bwm76j7yqcz3_s0cshbww80000gn/T/ipykernel_17379/3408759599.py:3: DeprecationWarning: executable_path
has been deprecated, please pass in a Service object
  driver = webdriver.Chrome('/Users/ejk/Downloads/chromedriver')

---------------------------------------------------------------------------
WebDriverException                        Traceback (most recent call last)
Input In [11], in <module>
      1 from selenium import webdriver
----> 3 driver = webdriver.Chrome('/Users/ejk/Downloads/chromedriver')

File /opt/homebrew/Caskroom/miniforge/base/envs/py39/lib/python3.9/site-packages/selenium/webdriver/chrome/webdriver.

    108 return_code = self.process.poll()
    109 if return_code:
--> 110     raise WebDriverException(
    111         'Service %s unexpectedly exited. Status code was: %s'
    112         % (self.path, return_code)
    113     )

WebDriverException: Message: Service /Users/ejk/Downloads/chromedriver unexpectedly exited. Status code was: -9
```

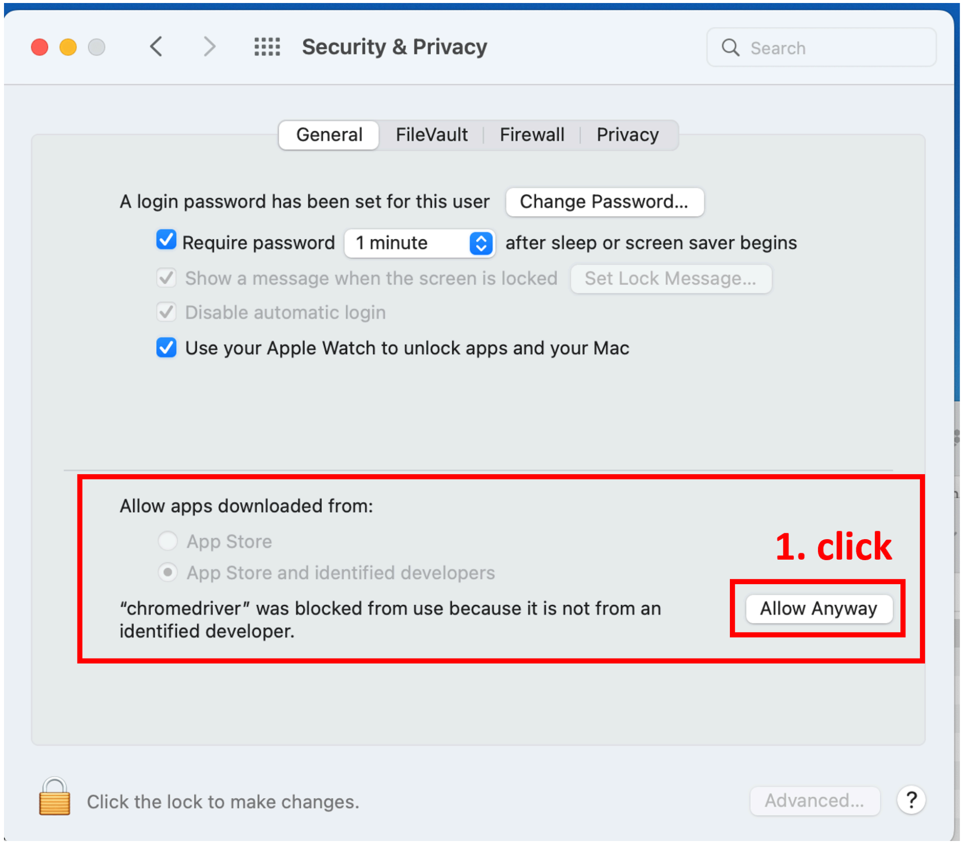
```
WebDriverException: Message: service /Users/... unexpectly exited. Status code was: -9
```

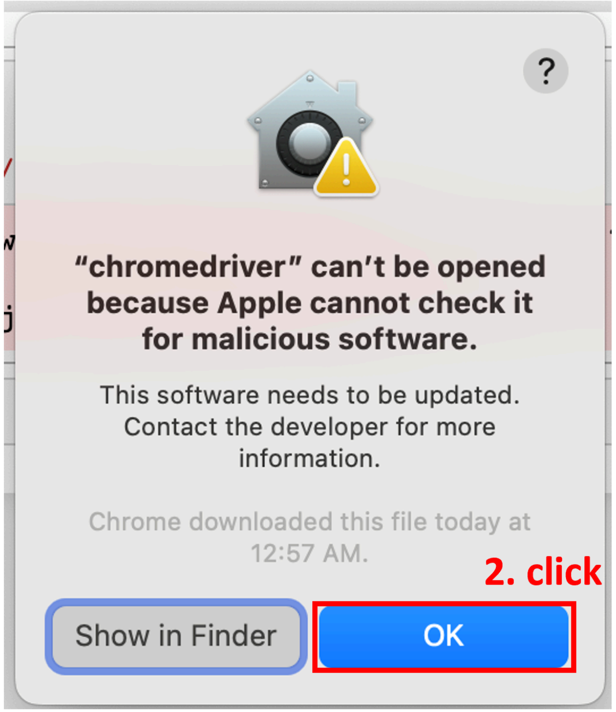
This is because MacOS prevented the Chrome driver from being opened.

Let's allow it to be opened!

How to fix it?

- `Security & Privacy setting` – `General tab`





Thank You.